A sharp positivity--rate frontier for quadratic fidelity of reversible spherical generators

Contents

Scope note. The frontier, equality, and stability theorems are valid in every ambient dimension `d>=2`. The matching constructions in Section 7 are proved only for `d=2,3`. No transport or physical-performance conclusion is made in this manuscript.

Abstract

Let $X = (\Omega_i)_{i\in I}$ be a finite indexed sample in $\mathbb{S}^{d-1}$ with positive masses w_i, and let L be a reversible Markov generator with nonnegative jump rates. Assume that constants and coordinate functions are reproduced exactly:

$$L\mathbf{1} = 0, \qquad L\Omega = -(d-1)\Omega.$$

We study the residual of L on sampled trace-free quadratic harmonics, quotienting exactly by the quadratic forms that vanish on X. If $r_{\max}$ is the largest outgoing rate and $\mathfrak{D}_2$ is the operator norm of this alias-correct residual in the sampled $\ell^2(w)$ metric, then

$$\mathfrak{D}_2 r_{\max} \geq d(d-1).$$

The constant is sharp in every dimension. We derive an exact covariance factorization into a scalar squared-loss component, with an exact rate-floor-plus-variance identity, and an orthogonal anisotropy component; we characterize equality by common rate and angular loss, with centered tight-frame conditions in every tangent space in the nonantipodal branch, a separate antipodal branch, and global reversible compatibility. We give regular-simplex, cross-polytope, and hypercube extremizers. We also derive an exact weighted decomposition of the frontier excess into radial and anisotropy defects, together with an explicit weighted concentration consequence. Stronger pointwise or global geometric conclusions require additional graph and nondegeneracy hypotheses and are not asserted as theorems here. The result is an operator frontier, not a quadrature-design statement or a generic obstruction to signed high-order discretizations. The order is attained by local positive families in dimensions two and three. Regular polygons give the exact product in $d = 2$. In $d = 3$, an explicit reflected adaptive-ring family has

$$r_{max}(L_h) \leq 64\pi^2 h^{-2}, \qquad \mathfrak{D}_2(L_h) \leq \frac{75}{2}h^2,$$

and therefore

$$\frac{3}{32\pi^2}h^2 \leq \inf_{L\in\mathcal{G}_h(64\pi^2)} \mathfrak{D}_2(L) \leq \frac{75}{2}h^2.$$

A perturbative robustness theorem for the three-dimensional construction is not claimed here. The fixed-support perturbation route is retained only as an open certification problem.

1. Introduction

Positive graph generators are attractive because their semigroups preserve order and mass. On a sampled sphere, exact reproduction of the coordinate functions is also natural: the coordinates form the first nonconstant Laplace–Beltrami eigenspace. These two requirements do not, however, leave arbitrary freedom at the next harmonic degree. This paper identifies the exact finite-dimensional obstruction.

Fix $d \geq 2$. We consider all finite positive reversible generators whose coordinate map is an eigenmap with eigenvalue $-(d-1)$. The next continuous spherical eigenvalue is $-2d$, so the relevant finite residual is

$$R_2 = (L + 2dI)S_2, \qquad (S_2A)_i = \Omega_i^{\mathsf{T}} A\Omega_i, \quad A \in \mathrm{Sym}_0(d).$$

The essential point is that S_2 need not be injective and its image need not be invariant under L. We therefore take the domain to be the sampled quotient $\mathrm{Sym}_0(d)/\ker S_2$, equipped with the norm induced by S_2, and allow

$R_2 A$ to have components outside $\operatorname{im} S_2$. This convention retains sampling aliases and leakage rather than silently discarding them.

The central theorem is the sharp product inequality

$$\boxed{\mathfrak{D}_2 r_{\max} \geq d(d-1)}$$

together with its equality and stability theory. The proof is finite and deterministic. It uses exact covariance identities forced by the coordinate eigenmap, positivity of the jump rates, and a trace comparison in the sampled metric.

1.1 Contributions

The paper makes the following mathematical contributions.

1. It gives an alias-correct formulation of sampled quadratic fidelity and an exact covariance representation of the residual.
2. It splits the rowwise residual into Frobenius-orthogonal scalar and anisotropic covariance components. The scalar squared-loss moment splits exactly into a rate floor and nonnegative angular-loss variance.
3. It proves the sharp all-dimensional frontier $\mathfrak{D}_2 r_{\max} \geq d(d-1)$, without injectivity of S_2, connectedness, equal masses, or invariance of the sampled quadratic space.
4. It characterizes equality both algebraically and geometrically. In the nonantipodal branch, each row induces a centered weighted unit-norm tight frame in the tangent space, while reversibility imposes additional global assembly conditions that are not consequences of rowwise feasibility.
5. It exhibits three exact extremal families in every dimension and resolves their sampling kernels explicitly.
6. It proves a normalized near-extremizer defect budget and a weighted exceptional-mass consequence. Stronger local and global transfers are discussed only as conditional directions requiring explicit mass, edge-probability, connectivity, sampling-frame, and tangent-frame margins.
7. It constructs matching-order local positive families in `d=2,3`. The circle family is an exact regular-polygon calculation. The spherical family uses shared conductances on adaptive reflected rings, proves every row moment equation at all levels, and gives the explicit constants $\mathsf{R}_3 = 64\pi^2$ and $C_3 = 75/2$.

1.2 Priority boundary

The contribution is not the introduction of positive coordinate-exact spherical Laplacians. Izmestiev and Lam already construct local reversible spherical Delaunay Laplacians on geodesic triangulations of $\mathbb{S}^2$, with nonnegative coefficients under their Delaunay condition and exact coordinate eigenvalue -2 (Izmestiev and Lam 2025). Positive meshfree stencils, graphical designs, prescribed graph eigenspaces, eigenpair-preserving Laplacian cones, spherical designs, association schemes, graph-Laplacian convergence, and signed higher-accuracy graph formulas likewise provide important adjacent theory (Seibold 2008; Babecki and Thomas 2022; Babecki and Shiroma 2023; Babecki et al. 2023; Bannai and Bannai 2009; Martin and Tanaka 2008; García Trillos et al. 2020; Yoon 2025).

The distinction here is the joint theorem: a sampled, alias-correct degree-two operator residual; a sharp universal positivity–rate product; its full equality geometry subject to global detailed balance; and a quantitative stability budget with all transfer parameters exposed. The matching three-dimensional construction additionally derives one globally shared positive stress on an explicit ring family along the discrete levels $J = 1, 2, \ldots$; it is not inferred from local positive-stencil feasibility, spherical quadrature, or a Delaunay transfer. A source-by-source hypothesis comparison and hostile-referee audit is supplied in PRIORITY_AND_HOSTILE_REFEREE_AUD IT.md. Within that focused corpus, we did not find the same joint package of frontier, equality characterization, exact defect identity, and local positive constructions. This is a positioning statement, not an exhaustive novelty or priority claim. Appendix G gives the in-paper variables/hypotheses/conclusions matrix for every compared theorem.

1.3 Organization

Section 2 fixes conventions and the sampled quotient. Section 3 proves the exact covariance and two-defect identities. Section 4 proves the sharp frontier. Section 5 characterizes equality and global assembly. Section 6 gives the exact defect budget and concentration consequence. Section 7 proves matching constructions in dimensions two and three. Section 8 gives exact extremizers and alias examples. Section 9 records false or overbroad statements ruled out by the theory. The appendices collect the dependency graph, provenance map, falsification protocol, formalization scope, reproducibility record, and limitations.

1.4 Theorem hierarchy

Level	Mathematical role	Principal statement	Status
1	Positive reversible generators and sampled harmonic quotients	Proposition 2.1	ordinary proof; finite Gram identities Lean-checked
2	Exact covariance and two-defect decomposition	Theorem 3.1 and Proposition 3.2	ordinary assembly; finite identities Lean-checked
3	Sharp positivity–rate–quadratic-fidelity frontier	Theorem 4.1	ordinary proof; lower-bound algebra Lean-checked
4	Equality geometry and exact extremizers	Theorems 5.1–5.3, Proposition 5.4, Corollaries 5.5–5.6, Section 8	ordinary proof; selected local algebra Lean-checked; Corollary 5.5 uses a classical external theorem
5	Quantitative near-extremizer control	Theorem 6.1 and Corollary 6.2	ordinary proof; stated scalar/tensor consequences Lean-checked
6	Matching-order local positive construction	Theorems 7.1–7.2	Theorem 7.1 ordinary/finite Lean algebra; Theorem 7.2 computer-assisted exact-rational proof for `d=3`
7	Generator-lift or negativity consequences	Section 10	deliberately not claimed

1.5 Formalization and external-input map

There is no claim that the complete article is formalized. The exact Lean declarations below are the checked finite components used by the ordinary proof; an entry reading "none end-to-end" means precisely that no single Lean theorem has the full manuscript signature.

Numbered result	Exact Lean declaration(s) used	Ordinary proof	Exact certificate	External or open boundary
Proposition 2.1	sphereSamplingWeightedGram__quadratic; sphereResidualWeightedGram__quadratic	yes	no	quotient/eigenvalue interpretation assembled in the article
Theorem 3.1	quadraticTwoDefect__pythagorean; exactLossSecondMoment__decomposition	yes	no	none
Proposition 3.2	weightedFrobeniusRowGram__pairing; sphereResidualWeightedGram__pairing	yes	no	none
Theorem 4.1	quadraticDefect__rate__product__lower; sTwo__quadraticDefect__rate__product__lower	yes	no	sharpness families are ordinary mathematics

Numbered result	Exact Lean declaration(s) used	Ordinary proof	Exact certificate	External or open boundary
Theorem 5.1	quadraticEquality_rate_product; equalityRemainderEntry_zero_iff_axialCovariance	yes	no	no end-to-end equality-classification declaration
Theorem 5.2	normalizedTangentTightFrame_iff_oneShellSecondMoment; scaledTangentTightFrame_iff_normalized_of_nonantipodal	yes	no	no global embedding uniqueness is asserted
Theorem 5.3	none end-to-end; local moment identities only	yes	no	Gram completion and Kolmogorov assembly are ordinary mathematics
Proposition 5.4	antipodalOnlyFeasible_iff_simplex is a local component	yes	no	connected two-position classification is not one Lean theorem
Corollary 5.5	normalizedTangentTightFrame_iff_oneShellSecondMoment is a local component	yes	no	convex regular-polyhedron classification is external (Coxeter 1973)
Corollary 5.6	detailedBalance_eigen_implies_weightedMean_zero; covariance components	yes	no	no end-to-end declaration
Theorem 6.1	quadraticStability_master_square_consequences; quadraticStability_tensor_rate_bound; quadraticStability_weightedMean_le_delta	yes	no	only the printed budget is claimed
Corollary 6.2	quadraticStability_badVertexMass_sharp; quadraticStability_pointwise_full_budget	yes	no	retains the individual mass dependence
Theorem 7.1	regularPolygon_hOne_scalar_recurrence; regularPolygon_hTwo_scalar_residual	yes	no	exact trigonometric assembly is ordinary mathematics
Theorem 7.2	construction identities only; no complete schedule Lean declaration	yes	certificate.json plus standalone verifier	computer-assisted analytic/Cauchy and recurrence boundary; perturbations open

2. Positive reversible spherical generators and sampled harmonic quotients

2.1 Notation

Symbol	Meaning	Convention or standing hypothesis
I	finite vertex set	nonempty
d	ambient Euclidean dimension	$d \geq 2$
Ω_i	spherical node	$\Omega_i \in \mathbb{S}^{d-1}$
w_i	stationary mass	$w_i > 0$, $\sum_i w_i = 1$
γ_{ij}	shared conductance	$\gamma_{ij} = \gamma_{ji} \geq 0$, $\gamma_{ii} = 0$
a_{ij}	directed jump rate	$a_{ij} = \gamma_{ij}/w_i$
L	negative Markov generator	$(Lf)_i = \sum_{j \neq i} a_{ij}(f_j - f_i)$
$r_i, r_{\max}$	outgoing rate and its maximum	$r_i = \sum_j a_{ij}$, $r_{\max} = \max_i r_i$
ℓ_{ij}	spherical loss	$1 - \Omega_i \cdot \Omega_j \in [0, 2]$
Δ_{ij}	chord increment	$\Omega_j - \Omega_i$

Symbol	Meaning	Convention or standing hypothesis
V	quadratic coefficient space	$\mathrm{Sym}_0(d)$
m	dimension of V	$d(d+1)/2-1$
S_2	quadratic sampling map	$(S_2A)_i = \Omega_i^{\mathsf{T}} A \Omega_i$
K_X	sampling-alias kernel	$\ker S_2 \subseteq V$
R_2	degree-two residual	$(L+2dI)S_2$
Q_X	sampled quotient	V/K_X with sampled metric
$\mathfrak{D}_2$	sampled quadratic defect	$\sup_{A\notin K_X} \lVert R_2A\rVert_w / \lVert S_2A\rVert_w$
C_i	row chord covariance	$\sum_j a_{ij}\Delta_{ij}\Delta_{ij}^{\mathsf{T}}$
Z_i	centered rank-one tensor	$\Omega_i\Omega_i^{\mathsf{T}} - I/d$
M_i	trace-free residual tensor	$C_i + 2\Omega_i\Omega_i^{\mathsf{T}} - 2I$
ϵ_i	squared-loss moment	$\sum_j a_{ij}\ell_{ij}^2$
B_i	anisotropy tensor	$M_i - \frac{d}{d-1}\epsilon_i Z_i$
E_ϵ, E_B	weighted defect energies	$\sum_i w_i\epsilon_i^2$, $\sum_i w_i\lVert B_i\rVert_F^2$

The weighted inner product is

$$\langle f, g\rangle_w = \sum_i w_i f_i g_i.$$

Detailed balance implies that L is self-adjoint and negative semidefinite in $\ell^2(w)$. Throughout,

$$L\mathbf{1} = 0, \qquad L\Omega = -(d-1)\Omega. \tag{2.1}$$

The second identity is componentwise in $\mathbb{R}^d$.

The quotient $Q_X = V/K_X$ carries the inner product

$$\langle [A], [H]\rangle_S = \langle S_2A, S_2H\rangle_w. \tag{2.2}$$

This is a genuine inner product on Q_X. It is not the quotient of the Frobenius metric unless an additional sampling-frame identity happens to hold.

Proposition 2.1 (sampled residual and aliases)

The inclusion $K_X \subseteq \ker R_2$ holds. Hence R_2 induces a well-defined map

$$\overline{R}_2 : Q_X \longrightarrow \ell^2(w), \qquad \overline{R}_2[A] = R_2A,$$

and

$$\mathfrak{D}_2 = \lVert \overline{R}_2 \rVert_{Q_X\to\ell^2(w)}. \tag{2.3}$$

Equivalently, $\mathfrak{D}_2^2$ is the largest finite generalized eigenvalue of the deflated Gram pencil

$$G_R = R_2^*R_2, \qquad G_S = S_2^*S_2$$

on the Frobenius complement $K_X^{\perp_F} \subseteq V$. No invariance of $\operatorname{im} S_2$ under L is required.

The genuinely sampled exact degree-two space is

$$E_{\text{sample}} = \operatorname{im} S_2 \cap \ker(L+2dI),$$

while the algebraic exact-form space is $E_{\text{form}} = \ker R_2 \subseteq V$. Then

$$\dim K_X = m - \operatorname{rank} S_2, \qquad \dim E_{\text{form}} = m - \operatorname{rank} R_2, \qquad \dim E_{\text{sample}} = \operatorname{rank} S_2 - \operatorname{rank} R_2. \tag{2.4}$$

The space E_{form} may contain the alias kernel K_X; only E_{sample} records genuinely sampled exact modes.

3. Exact covariance and the two-defect decomposition

The coordinate eigenmap fixes the first loss moment in every row:

$$\sum_j a_{ij}\ell_{ij} = d-1. \tag{3.1}$$

This elementary identity is the point at which the continuous coordinate eigenvalue enters all subsequent estimates.

Theorem 3.1 (exact covariance representation)

For every $A \in \mathrm{Sym}_0(d)$ and every vertex i,

$$(R_2A)_i = \langle A, M_i\rangle_F. \tag{3.2}$$

Moreover,

$$\langle M_i, Z_i\rangle_F = \epsilon_i, \qquad \|Z_i\|_F^2 = \frac{d-1}{d}, \qquad \langle B_i, Z_i\rangle_F = 0, \tag{3.3}$$

and therefore

$$M_i = \frac{d}{d-1}\epsilon_i Z_i + B_i, \qquad \|M_i\|_F^2 = \frac{d}{d-1}\epsilon_i^2 + \|B_i\|_F^2. \tag{3.4}$$

If

$$V_i = \sum_j a_{ij}\left(\ell_{ij} - \frac{d-1}{r_i}\right)^2,$$

then

$$\epsilon_i = \frac{(d-1)^2}{r_i} + V_i. \tag{3.5}$$

Thus the scalar part of the quadratic residual is the unavoidable rate term $(d-1)^2/r_i$ plus active-loss variance V_i, while B_i is the orthogonal covariance-anisotropy part.

Proposition 3.2 (weighted adjoints and exact traces)

Use the Frobenius inner product on V and the weighted inner product on samples. The adjoints are

$$S_2^* f = \sum_i w_i f_i Z_i, \qquad R_2^* f = \sum_i w_i f_i M_i. \tag{3.6}$$

Consequently,

$$\operatorname{tr} G_S = \frac{d-1}{d}, \qquad \operatorname{tr} G_R = \frac{d}{d-1}E_\epsilon + E_B. \tag{3.7}$$

These trace identities use the sampled metric and remain valid when S_2 has a nontrivial kernel.

Proof architecture

Equation (3.2) follows by expanding each sampled quadratic along the chord Δ_{ij} and applying (2.1). Equation (3.3) follows by contracting with Z_i, after which (3.4) is an orthogonal projection. Equation (3.5) is the weighted mean–variance identity with fixed mean (3.1). Finally, (3.6)–(3.7) follow by weighted duality and the row-frame trace formula. Complete proofs are assembled from the P1A source identified in Appendix B.

4. The sharp positivity–rate–quadratic-fidelity frontier

Theorem 4.1 (sharp trace and product bounds)

Every generator satisfying the hypotheses of Section 2 obeys

$$\mathfrak{D}_2^2 \geq \frac{d^2}{(d-1)^2}E_\epsilon + \frac{d}{d-1}E_B. \tag{4.1}$$

In particular,

$$\mathfrak{D}_2 \geq \frac{d}{d-1}\sqrt{E_\epsilon} \geq \frac{d(d-1)}{r_{\max}}, \tag{4.2}$$

and hence

$$\boxed{\mathfrak{D}_2 r_{\max} \geq d(d-1).} \tag{4.3}$$

The constant is attained for every $d \geq 2$.

Proof architecture

On $K_X^{\perp_F}$, compare the trace of the positive Gram operator G_R with $\mathfrak{D}_2^2 G_S$. The exact traces (3.7) give (4.1). Next use (3.5), $V_i \geq 0$, $r_i \leq r_{\max}$, and $\sum_i w_i = 1$ to obtain (4.2). The regular-simplex family in Section 8 attains equality, so the constant cannot be improved.

Remark 4.2 (scope of the frontier)

The theorem is neither a general no-go theorem for high-order graph Laplacians nor a statement about quadrature exactness. It applies to the specified class of nonnegative reversible generators, with exact coordinate eigenvalue, the sampled degree-two quotient norm, and the maximum outgoing rate. Signed formulas and other consistency norms lie outside its hypotheses.

5. Equality geometry and exact extremizers

Theorem 5.1 (algebraic equality characterization)

Equality holds in (4.3) if and only if, for every $i \in I$,

$$r_i = r_{\max}, \qquad V_i = 0, \qquad B_i = 0. \tag{5.1}$$

Equivalently, with

$$c_* = \frac{d(d-1)}{r_{\max}},$$

one has

$$\epsilon_i = \frac{(d-1)^2}{r_{\max}}, \qquad M_i = c_* Z_i \quad \text{for every } i, \tag{5.2}$$

and

$$R_2 = c_* S_2 \tag{5.3}$$

on the full coefficient space and therefore on the sampled quotient.

At equality,

$$\ker R_2 = \ker S_2 = K_X, \qquad E_{\text{sample}} = \{0\}. \tag{5.4}$$

Thus equality does not reproduce the continuous degree-two eigenvalue; it produces the smallest positive residual allowed by positivity and the rate cap.

5.1 Tangent decomposition

Fix i, write $u = \Omega_i$, and let

$$P_i = I - uu^\mathsf{T}, \qquad \tau_{ij} = P_i \Omega_j.$$

Then

$$\Delta_{ij} = -\ell_{ij} u + \tau_{ij}, \qquad \|\tau_{ij}\|^2 = \ell_{ij}(2-\ell_{ij}), \qquad \sum_j a_{ij}\tau_{ij} = 0. \tag{5.5}$$

Define

$$h_i = \sum_j a_{ij}\ell_{ij}\tau_{ij}, \qquad T_i = \sum_j a_{ij}\tau_{ij}\tau_{ij}^\mathsf{T}.$$

Theorem 5.2 (local block identity and tight-frame form)

The anisotropy tensor has the exact orthogonal block form

$$B_i = -uh_i^\mathsf{T} - h_i u^\mathsf{T} + T_i - \left(2 - \frac{\epsilon_i}{d-1}\right) P_i, \tag{5.6}$$

and

$$\|B_i\|_F^2 = 2\|h_i\|^2 + \left\|T_i - \left(2 - \frac{\epsilon_i}{d-1}\right) P_i\right\|_F^2. \tag{5.7}$$

If $V_i = 0$, all active edges in row i have the common loss

$$\bar{\ell}_i = \frac{d-1}{r_i}.$$

In the nonantipodal case $0 < \bar{\ell}_i < 2$, let

$$p_{ij} = \frac{a_{ij}}{r_i}, \qquad y_{ij} = \frac{\tau_{ij}}{\sqrt{\bar{\ell}_i(2-\bar{\ell}_i)}}.$$

Then $B_i = 0$ if and only if

$$\sum_j p_{ij} y_{ij} = 0, \qquad \sum_j p_{ij} y_{ij} y_{ij}^\mathsf{T} = \frac{1}{d-1} P_i. \tag{5.8}$$

Thus the active tangent directions form a centered weighted unit-norm tight frame. The antipodal case $\bar{\ell}_i = 2$ is governed by the division-free identities (5.6)–(5.7), not by the normalization in (5.8).

Theorem 5.3 (global reversible assembly)

In the nonantipodal branch, put

$$t = 1 - \ell_* \in (-1, 1), \qquad p_{ij} = a_{ij}/r_*.$$

Frontier equality is equivalent to one common outgoing rate $r_* = (d-1)/(1-t)$ together with

$$\sum_j p_{ij} = 1, \qquad w_i p_{ij} = w_j p_{ji}, \qquad p_{ij} > 0 \Longrightarrow \Omega_i \cdot \Omega_j = t, \tag{5.9}$$

$$\sum_j p_{ij} \Omega_j = t\Omega_i, \tag{5.10}$$

and

$$\sum_j p_{ij} \Omega_j \Omega_j^\mathsf{T} = t^2 \Omega_i \Omega_i^\mathsf{T} + \frac{1-t^2}{d-1}(I - \Omega_i \Omega_i^\mathsf{T}) \tag{5.11}$$

for every vertex i. Conversely, (5.9)–(5.11), with $a_{ij} = r_* p_{ij}$, reconstruct the coordinate eigenmap and all equality conditions.

Equivalently, retaining the Markov and detailed-balance conditions in (5.9), the Gram formulation requires a single matrix $G \succeq 0$ with $G_{ii} = 1$, $\operatorname{rank} G = d$, and $p_{ij} > 0 \Rightarrow G_{ij} = t$, together with

$$\sum_j p_{ij} G_{jk} = t G_{ik} \tag{5.12}$$

and, for all i, k, l,

$$\sum_j p_{ij} G_{jk} G_{jl} = \alpha G_{kl} + (t^2 - \alpha) G_{ik} G_{il}, \qquad \alpha = \frac{1-t^2}{d-1}. \tag{5.13}$$

On a connected bidirected support, positive reversible masses exist exactly when the directed rates satisfy the Kolmogorov identity on every oriented cycle. In particular, local row feasibility does not imply the existence of a global reversible spherical generator.

Proposition 5.4 (connected antipodal branch)

If the common equality loss is $\ell_* = 2$ and the active graph is connected, all indexed nodes occupy two antipodal positions $\pm u$, every active edge joins opposite positions, and

$$r_* = \frac{d-1}{2}, \qquad R_2 = 2dS_2. \tag{5.14}$$

Conversely, every reversible connected constant-row-rate generator with this geometry is an equality generator. Repeated indices at either antipode are allowed.

Corollary 5.5 (degree and classification consequences)

Every nonantipodal equality row has at least d active neighbors. If it has exactly d, its normalized tangent directions form a regular simplex with equal weights. A distinct nonantipodal complete-support equality configuration is the ambient regular simplex.

For $d = 3$, assume additionally that $0 < \ell_* < 2$, the nodes are the distinct vertices of a strictly convex inscribed polyhedron, the active graph is exactly its one-skeleton, all active directed rates have one common value, and the common vertex degree q satisfies $3 \le q \le 5$. Then the local equations force a regular vertex figure and the elementary face/Euler reduction gives one of the five pairs (p, q). The classical classification theorem for convex regular polyhedra then implies that the configuration is tetrahedral, octahedral, cubical, icosahedral, or dodecahedral (Coxeter 1973). This last classification input is external to Lean. No corresponding classification is asserted for general weighted equality configurations.

Corollary 5.6 (global moment consequence)

Every nonantipodal equality generator satisfies

$$\sum_i w_i \Omega_i = 0, \qquad \sum_i w_i \Omega_i \Omega_i^{\mathsf{T}} = \frac{1}{d} I. \tag{5.15}$$

Thus its stationary node measure is a weighted spherical 2-design. This is a consequence of the reversible equality system; it is neither a standing design hypothesis nor an identification of operator fidelity with quadrature exactness.

6. Quantitative near-extremizer stability

Assume

$$\mathfrak{D}_2 r_{\max} \le d(d-1)(1+\delta), \qquad \delta \ge 0. \tag{6.1}$$

Set

$$n = d-1, \qquad a_0 = \frac{n^2}{r_{\max}}, \qquad c_0 = \frac{dn}{r_{\max}}, \qquad \eta = 2\delta + \delta^2,$$

and define

$$x_i = \frac{\epsilon_i}{a_0}, \qquad s_i = \frac{r_{\max}}{r_i} - 1, \qquad v_i = \frac{V_i}{a_0}, \qquad q_i = x_i - 1 = s_i + v_i. \tag{6.2}$$

The defects s_i, v_i, q_i are nonnegative, and $x_i = 1 + q_i \ge 1$.

Theorem 6.1 (master stability budget)

Under (6.1),

$$\sum_i w_i (2q_i + q_i^2) + \frac{d-1}{da_0^2} \sum_i w_i \|B_i\|_F^2 \le \eta. \tag{6.3}$$

Consequently,

$$\sum_i w_i (x_i - 1)^2 \le \eta, \qquad \sum_i w_i \left(\frac{r_{\max}}{r_i} - 1\right)^2 \le \eta, \tag{6.4}$$

$$\sum_i w_i \|B_i\|_F^2 \le \frac{d(d-1)^3}{r_{\max}^2}\eta, \tag{6.5}$$

and

$$\sum_i w_i q_i \le \delta, \quad \sum_i w_i s_i \le \delta, \quad \sum_i w_i v_i \le \delta, \tag{6.6}$$

$$\sum_i w_i V_i \le a_0 \delta, \qquad \sum_i w_i V_i^2 \le a_0^2 \eta. \tag{6.7}$$

The estimates are valid without connectedness, injectivity of S_2, equal masses, or a uniform lower mass bound.

Corollary 6.2 (exceptional mass and pointwise transfer)

Let

$$H = \eta - \frac{d-1}{da_0^2} \sum_i w_i \|B_i\|_F^2.$$

Then $0 \le H \le \eta$ and $\sum_i w_i(2q_i + q_i^2) \le H$. For every $\rho > 0$,

$$\sum_{i: q_i \ge \rho} w_i \le \min\left\{1, \frac{H}{\rho(2+\rho)}\right\}. \tag{6.8}$$

At an individual vertex,

$$q_i \le \sqrt{1 + \frac{H}{w_i}} - 1. \tag{6.9}$$

Thus a uniform pointwise conclusion requires an explicit lower bound on the stationary masses.

Conditional transfers from the defect budget (discussion)

Theorem 6.1 gives weighted mean-square control of radial nonuniformity and local anisotropy. Pointwise, edgewise, graphwise, quotient, and frame conclusions require additional assumptions, such as lower bounds on node or edge weights, a quantitative spectral gap or diameter bound, and nondegeneracy of reference tangent frames. No unconditional global embedding-stability theorem is asserted here. The following list records what a separately stated transfer theorem would have to retain; the list is not a numbered corollary and is not itself a theorem.

- The global shell loss has an explicit second-moment bound in the symmetric directed conductance measure. To control every active edge, one assumes $p_{ij} = a_{ij}/r_i \ge \kappa > 0$; the pointwise bound still displays w_i, and a uniform version requires $w_{\min} > 0$.
- Graphwise propagation continues the hypotheses $p_{ij} \ge \kappa > 0$ and $w_i \ge w_{\min} > 0$ on the connected support. Multiplicative path bounds require the stated small-defect condition; additive path, effective-resistance, Poincaré-gap, and congestion variants retain their corresponding graph parameter.
- The orthogonal block identity (5.7) converts the tensor term of the master budget into weighted control of mixed radial–tangent covariance and tangent anisotropy. Together with the scalar term, it controls the full covariance relative to its stated target and its eigenvalue splitting. Eigenvector control further requires a spectral gap.
- For example, let $W_X = K_X^{\perp_F}$, $E = \operatorname{im} S_2$, and

  $$\alpha_X = \inf_{\substack{A \in W_X \\ \|A\|_F = 1}} \|S_2 A\|_w^2 > 0.$$

  Define $U : E \to \ell^2(w)$ by $U(S_2 A) = R_2 A$, and let $\iota_E : E \hookrightarrow \ell^2(w)$ denote inclusion. The lower sampling-frame bound converts the full master budget into quotient-correct operator scalarity, including leakage:

  $$\|U - c_0 \iota_E\|_{\mathrm{op}} \le \sqrt{\frac{K_0}{\alpha_X}}, \qquad K_0 = \frac{d(d-1)^3}{r_{\max}^2}\eta. \tag{6.10}$$

- A tangent lower-frame bound gives rotation-quotiented distance to an exact centered weighted tight frame. A positive unit-frame and spherical-neighbor repair further requires nonantipodal shell margins, a lower active probability, and a feature-surjectivity bound.

No parameter-free pointwise, edgewise, graphwise, quotient, or frame-repair statement is inferred from (6.3), and no claim in this discussion may be cited as a proved geometric-stability theorem.

Proof architecture

Insert the near-frontier assumption into the trace inequality (4.1), normalize by a_0, and use $x_i = 1 + q_i$. This gives (6.3) exactly. The global consequences follow by positivity and elementary moment inequalities. The subsequent discussion is deliberately non-theorem scope; the counterexamples in Section 9 show why its additional margins cannot simply be omitted.

7. Matching-order positive constructions in dimensions two and three

Write

$$d_{\mathbb{S}}(x, y) = \arccos(x \cdot y), \quad h_{\text{fill}}(X) = \sup_{x \in \mathbb{S}^{d-1}} \min_i d_{\mathbb{S}}(x, \Omega_i),$$

$$q_{\text{sep}}(X) = \frac{1}{2} \min_{i \neq j} d_{\mathbb{S}}(\Omega_i, \Omega_j), \qquad \rho_X = \frac{h_{\text{fill}}(X)}{q_{\text{sep}}(X)}.$$

For a mesh X_h, let $\mathcal{G}_h(\mathsf{R})$ denote the positive reversible generators on X_h that reproduce constants and coordinates exactly and satisfy $r_{\max} \leq \mathsf{R} h^{-2}$. Any further support restriction will be stated with the construction. The lower half of each estimate below is Theorem 4.1; the work in this section is the matching upper half.

Theorem 7.1 (regular polygons on $\mathbb{S}^1$)

Let $N \geq 5$, put $h = \pi/N$, and set

$$\Omega_k = (\cos(2kh), \sin(2kh)), \qquad k \in \mathbb{Z}/N\mathbb{Z}.$$

Then $h_{\text{fill}}(X_h) = q_{\text{sep}}(X_h) = h$ and $\rho_{X_h} = 1$. Take $w_k = 1/N$ and the two nearest-neighbor rates

$$a_{k,k-1} = a_{k,k+1} = \frac{1}{2\{1 - \cos(2h)\}}. \tag{7.1}$$

Then the graph has degree two and active edge angle $2h$, and

$$L\mathbf{1} = 0, \qquad L\Omega = -\Omega,$$

$$r_{\max} = \frac{1}{1 - \cos(2h)}, \qquad \mathfrak{D}_2 = 2\{1 - \cos(2h)\} = 4\sin^2 h, \tag{7.2}$$

so $\mathfrak{D}_2 r_{\max} = 2$. In particular, with

$$\mathsf{R}_2 = \frac{\pi^2}{8}, \qquad C_2 = 4, \qquad c_2 = \frac{16}{\pi^2},$$

one has

$$\frac{16}{\pi^2} h^2 \leq \inf_{L \in \mathcal{G}_h(\mathsf{R}_2)} \mathfrak{D}_2(L) \leq 4h^2. \tag{7.3}$$

Proof The nearest-neighbor sum sends the frequency-m Fourier mode to $2(\cos(2mh) - 1)$ times that mode. With (7.1), the frequency-one eigenvalue is -1, proving exact coordinate reproduction. The sampled trace-free quadratics are the frequency-two sine and cosine modes; $N \geq 5$ makes their sampled space two-dimensional. Their generator eigenvalue is

$$-\frac{1 - \cos(4h)}{1 - \cos(2h)} = -2\{1 + \cos(2h)\}.$$

Adding the continuous degree-two eigenvalue 4 gives the scalar residual in (7.2). For $0 < h \le \pi/5$,

$$\frac{2h}{\pi} \le \sin h \le h,$$

and hence $1 - \cos(2h) = 2\sin^2 h \ge 8h^2/\pi^2$. This proves the rate cap and upper bound. Theorem 4.1 gives $\mathfrak{D}_2 \ge 2/r_{\max} \ge 2h^2/\mathsf{R}_2$, which is the lower bound in (7.3). □

7.2 The reflected adaptive-ring construction on $\mathbb{S}^2$

We give the construction data before stating the theorem. Fix

$$M_0 = 2^{80}, \qquad a_{\text{pole}} = \frac{4}{3}, \qquad g = \sqrt{\frac{29}{32}}. \tag{7.4}$$

At level $J \ge 1$, put $M_m = M_0 2^m$ and $S_0 = M_J/8$. For $0 \le m < J$ define

$$t_m^0 = \frac{M_J}{4\pi}\arcsin(2^{m-J}), \qquad k_m = \operatorname{nint}(t_m^0 - a_{\text{pole}} - mg),$$

and put

$$K = \operatorname{nint}(S_0 - a_{\text{pole}} - Jg), \quad t_m = a_{\text{pole}} + k_m + mg, \quad S = a_{\text{pole}} + K + Jg, \quad h = \frac{\pi}{2S}. \tag{7.5}$$

Nearest-integer ties are broken downward. Starting with the north pole and a first ring at latitude $a_{\text{pole}}h$, insert k_0 gaps of size h, one gap of size gh at which the longitude count doubles, then $k_{m+1} - k_m$ ordinary gaps between successive doublings, and finally $K - k_{J-1}$ ordinary gaps before the equator. Reflect through the equator without duplicating the equatorial ring. A ring in the mth band has M_m equally spaced longitudes. The rounding identities give

$$|t_m - t_m^0| \le \frac{1}{2}, \qquad |S - S_0| \le \frac{1}{2}, \qquad \left|\frac{2S}{M_m}\sin\frac{\pi t_m}{2S} - \frac{1}{4}\right| \le \frac{3}{M_m}. \tag{7.6}$$

Aligned rings use identity longitude matching. At a doubling interface $M : 2M$, put $\alpha = \pi/M$, $c_\alpha = \cos\alpha$, and

$$p = \frac{4c_\alpha^2 + 2c_\alpha - 1}{4c_\alpha(c_\alpha + 1)}, \qquad q = \frac{(2c_\alpha + 1)(4c_\alpha^2 + 2c_\alpha - 1)}{8c_\alpha^2(c_\alpha + 1)}. \tag{7.7}$$

For coarse longitude i and fine offset $r = j - 2i$, the seven nonzero entries of the shared radial mask are

$$\begin{array}{c|ccccccc} r & 0 & 2 & -2 & 1 & -1 & 3 & -3 \\ \hline Q_{i,2i+r} & p/2 & (1-p)/4 & (1-p)/4 & q/4 & q/4 & (1-q)/4 & (1-q)/4. \end{array} \tag{7.8}$$

Its row sum is one, every column sum is one half, and its reverse even/odd conditional laws have the same first two cosine moments.

Preliminary symmetric stresses $\Gamma_{ij} = \Gamma_{ji}$ are fixed recursively from the pole. At every ordinary row, two reflected horizontal pairs bracket the target loss by the integer rule

$$t_- = \max\{1, \lfloor \phi_*/(2\delta) \rfloor\}, \qquad t_+ = \lceil 2\phi_*/\delta \rceil, \tag{7.9}$$

where $\delta = 2\pi/M$ and

$$\sin(\phi_*/2) = \frac{\sin(h/2)}{\sin\theta}.$$

Given the incoming radial stress, the outgoing radial stress and the two horizontal stresses are the unique solution of the three even moment equations

$$\sum_j \Gamma_{ij}(\Delta_{ij} \cdot e_\theta) = 0, \qquad \sum_j \Gamma_{ij}\ell_{ij}(\Delta_{ij} \cdot e_\theta) = 0, \tag{7.10}$$

$$\sum_j \Gamma_{ij}\left\{(\Delta_{ij} \cdot e_\theta)^2 - (\Delta_{ij} \cdot e_\phi)^2\right\} = 0. \tag{7.11}$$

At a doubling, the two endpoint rows and their shared radial stress are solved simultaneously in the corresponding 6×6 system using (7.8). For the first ring, put $\delta_0 = 2\pi/M_0$ and choose the two horizontal jump integers nearest to $(3/8)/\delta_0$ and $(9/8)/\delta_0$, with the same downward tie rule. Set every pole-to-first-ring preliminary stress equal to 1, then solve the resulting 3×3 system for the outgoing radial and two horizontal stresses. At the equator use equal reflected incoming stresses and horizontal jump two; reflection cancels (7.10), and the positive horizontal stress enforces (7.11). Longitude reflection cancels every odd-e_ϕ equation.

The exact transition matrix has a removable $M^{-1} = 0$ limit whose inverse has norm below 3 and whose solution has minimum greater than 1/500. On the correlated box from (7.6), a rational Cauchy estimate gives

$$\|A - A_0\|_{\max} \le \frac{10^{12}}{M}, \qquad \|b - b_0\|_\infty \le \frac{10^{12}}{M}. \tag{7.12}$$

Here is the load-bearing transition certificate explicitly. After the row scalings specified by (7.10)–(7.11), write $x = M^{-1}$, $y = \eta/(\pi c) \in [-7/6, 7/6]$, and $\sigma = \sqrt{58}$. The removable system $A(x)z = b(x)$ has

$$A_0 = \begin{pmatrix} -29/8 & \sigma/8 & 8\sigma & 0 & 0 & 0 \\ -29/8 & \sigma/32 & 128\sigma & 0 & 0 & 0 \\ 0 & \sigma/8 & 8\sigma & 0 & 0 & 0 \\ 29/8 & 0 & 0 & -\sigma & \sigma/16 & 4\sigma \\ 29/8 & 0 & 0 & -\sigma & \sigma/256 & 16\sigma \\ 0 & 0 & 0 & 0 & \sigma/16 & 4\sigma \end{pmatrix}, \tag{7.12a}$$

$$b_0(y) = \begin{pmatrix} -3/16 \\ 63/512 + 3\sigma y/32 \\ 13/8 + \sigma/4 \\ -3/16 \\ -285/512 - 3\sigma y/32 \\ 13/8 + \sigma/4 \end{pmatrix}. \tag{7.12b}$$

Exact arithmetic in $\mathbb{Q}(\sqrt{58})$ gives

$$\det A_0 = \frac{96799941}{512}\sqrt{58}, \qquad \|A_0^{-1}\|_\infty < 3, \qquad \frac{1}{500} < (A_0^{-1} b_0(y))_k < 10. \tag{7.12c}$$

The Cauchy disk is $|x| \le 10^{-4}$, with the rational guards $a > 1/5$, $c > 6/7$, and $\cos(\pi x) > 9/10$. On that disk every normalized matrix entry is below 10^6 and every right-hand entry is below 10^7. Cauchy's estimate yields the deliberately weaker bounds (7.12). Since

$$18\frac{10^{12}}{2^{80}} < 2^{-35}, \qquad 244\frac{10^{12}}{2^{80}} < \frac{1}{1000} < \frac{1}{500}, \tag{7.12d}$$

the Neumann series preserves strict positivity throughout the correlated transition box. These are exact rational comparisons, not sampled or floating-point estimates.

For the separate first polar row, $a = 4/3$, $u \in (1/20, 1/10)$ and $v \in (1/2, 2/3)$. Its normalized limiting determinant is $4a^3uv(a-1)(2a+1)(u-v)$, whose absolute value is at least 704/6075. The polar inverse has norm below 20,000; the rational derivative bounds $\|\partial_h A_{\mathrm{pol}}\|_{\max} \le 22{,}000$ and $\|\partial_h b_{\mathrm{pol}}\|_\infty \le 1{,}600$, with $h \le 44/(7 \cdot 2^{80})$, make its finite solution differ from its positive limiting solution by less than 10^{-6}. For ordinary rows, (7.9) gives the uniform bracket

$$\frac{1}{16} < x_- < \frac{1}{3}, \qquad 2 < x_+ < 7, \qquad \frac{1}{100}U_- < H_-, H_+ < 100U_-. \tag{7.13}$$

For completeness, if $z = \tan(h/2)\cot\theta$, the exact shared-edge recurrence on an ordinary row is

$$\frac{U_+}{U_-} = \frac{(1-z)(1+2z)}{(1+z)(1-2z)}, \qquad 0 \le E(z) \le \frac{16z^3}{1-4z^2}, \tag{7.13a}$$

where $\log(U_+/U_-) = \log\{\sin(\theta + h/2)/\sin(\theta - h/2)\} + E(z)$. The sine quotient telescopes. The cap error sums to less than 12, the dyadic-band errors sum to less than 1, and the mth count change has $|\varepsilon_m| < 256/M_m$; hence its cumulative product lies between e^{-512/M_0} and e^{512/M_0}. The exact telescoping recurrence, including all count changes, therefore yields the level-independent shared-stress margins

$$\Gamma_- = M_0^{-20} \le \Gamma_{ij} \le M_0^{20} = \Gamma_+. \tag{7.14}$$

Thus positivity is certified before any normalization; it is not inferred from the finite generator.

Computer-assisted proof boundary. The analytic reduction, literal systems, recurrence, and geometric normalization are printed here and in the mathematical supplement P1E_SHORT_GAP_S2_CONSTRUCTION.md. The remaining finite rational inequalities are recorded in the immutable artifact afp_barrier_gate1/release/certificates/theorem_7_2/certificate.json and checked by the standalone standard-library program verify_certificate.py. The verifier recomputes the determinant, inverse and affine solution in $\mathbb{Q}(\sqrt{58})$, all Cauchy/Neumann and polar-row inequalities, source hashes, recurrence closure, and the published constants without floating point or third-party imports. The trusted computing base is the certificate and verifier bytes, CPython arbitrary-precision integer/Fraction arithmetic, SHA-256, and the stated analytic estimates. The adaptive-ring schedule and Cauchy argument are not Lean-checked. Theorem 7.2 is therefore explicitly a computer-assisted exact-rational theorem.

Theorem 7.2 (matching order on $\mathbb{S}^2$)

For every integer $J \ge 1$, let $h_J = \pi/(2S_J)$ be the discrete scale defined by (7.4)–(7.5). For this explicit sequence, the family (7.4)–(7.14) has positive weights and a local reversible generator with

$$L_h\mathbf{1} = 0, \qquad L_h\Omega = -2\Omega, \qquad \gamma_{ij} = \gamma_{ji} \ge 0. \tag{7.15}$$

The metrics defined above satisfy

$$h_{\mathrm{fill}}(X_h) \le 2h, \qquad q_{\mathrm{sep}}(X_h) \ge q_* h, \qquad \rho_{X_h} \le \frac{2}{q_*}, \qquad q_* = (8M_0)^{-1}.$$

Every active edge lies in the angular window $h/8 \le d_{\mathbb{S}}(\Omega_i, \Omega_j) \le 5h$, and the degree is at most $D = M_0$. For every vertex,

$$B_i = 0, \qquad (R_2A)_i = c_i(S_2A)_i, \qquad 0 < c_i \le \frac{75}{2}h^2. \tag{7.16}$$

Consequently,

$$r_{\max}(L_h) \le 64\pi^2 h^{-2}, \qquad \mathfrak{D}_2(L_h) \le \frac{75}{2}h^2. \tag{7.17}$$

With

$$\mathsf{R}_3 = 64\pi^2, \qquad C_3 = \frac{75}{2}, \qquad c_3 = \frac{3}{32\pi^2},$$

one therefore has, for every $J \ge 1$ at the corresponding scale h_J,

$$\frac{3}{32\pi^2}h^2 \le \inf_{L\in\mathcal{G}_h(\mathsf{R}_3)} \mathfrak{D}_2(L) \le \frac{75}{2}h^2. \tag{7.18}$$

Proof The schedule closes exactly because the terminal dimensionless latitude is $a_{\mathrm{pole}} + k_{J-1} + (J-1)g + g + K - k_{J-1} = S$. Equations (7.6), the ring spacing, and the active jumps prove the stated fill, separation, window, degree, and packing bounds. The exact row systems (7.10)–(7.11), the shared column sum in (7.8), the Cauchy bounds (7.12), the ordinary margins (7.13), and the telescoping product (7.14) prove positivity and assign each undirected edge one preliminary symmetric stress.

Set

$$\mu_i = \frac{1}{2}\sum_j \Gamma_{ij}\ell_{ij}, \qquad W = \sum_i \mu_i, \qquad w_i = \frac{\mu_i}{W}, \qquad \gamma_{ij} = \frac{\Gamma_{ij}}{W}. \tag{7.19}$$

The tangent force equation and radial contraction give

$$\sum_j \Gamma_{ij}\Delta_{ij} = -2\mu_i\Omega_i.$$

Therefore $a_{ij} = \gamma_{ij}/w_i = \Gamma_{ij}/\mu_i$ proves (7.15). Let $R_i = \sum_j \Gamma_{ij}\ell_{ij}^2$. The loss-force and isotropy equations give, in the radial–tangent frame,

$$M_i = \frac{1}{\mu_i}\operatorname{diag}(R_i, -R_i/2, -R_i/2) = \frac{3R_i}{2\mu_i} Z_i. \tag{7.20}$$

This proves $B_i = 0$ and (7.16) with $c_i = 3R_i/(2\mu_i)$. Since every active edge has angle at most $5h$,

$$c_i \le 3\ell_{\max} \le \frac{75}{2}h^2.$$

The row multiplier descends directly to the sampled quotient: if $S_2A = 0$, then (7.16) gives $R_2A = 0$. Thus no sampling-frame denominator occurs.

Every active edge has loss at least $h^2/(32\pi^2)$, so

$$r_i = \frac{2\sum_j \Gamma_{ij}}{\sum_j \Gamma_{ij}\ell_{ij}} \le 64\pi^2 h^{-2},$$

which proves (7.17). The lower half of (7.18) is Theorem 4.1: $\mathfrak{D}_2 \ge 6/r_{\max} \ge 6h^2/\mathsf{R}_3$. $\square$

Open problem 7.3 (support-preserving perturbations)

The fixed-support reflected-latitude perturbation route remains plausible, but it is **not** a theorem of this paper. The repository does not contain the literal cancellation-free straight-line program, verified operation count, complete denominator-guard inventory, or machine-checkable derivative and global-recurrence certificate needed to justify the previously advertised universal constant. The associated scripts are retained as diagnostics and falsification aids only.

A complete robustness theorem would have to preserve the ring counts, longitude phases, radial masks, horizontal jump integers, pole and equator, and north–south reflection; enumerate the exact normalized row programs; certify every reciprocal guard and derivative bound; and propagate the local bounds through the shared-conductance recurrence. Until such a certificate is committed and independently checked, no perturbation radius or constants $\mathsf{R}_{\mathrm{rob}}$ and C_{rob} are asserted.

8. Exact examples

8.1 Three extremal families in every dimension

All entries below are exact. Every family has uniform stationary masses, nonnegative reversible rates, exact coordinate eigenvalue $-(d-1)$, and $\mathfrak{D}_2 r_{\max} = d(d-1)$.

Family	Nodes and active graph	Directed active rate	$r_{\max}$	Common loss	$\mathfrak{D}_2$	$\operatorname{rank} S_2$	$\dim K_X$
Regular simplex	$d+1$ simplex vertices; complete graph	$(d-1)/(d+1)$	$d(d-1)/(d+1)$	$(d+1)/d$	$d+1$	d	$(d+1)(d-2)/2$
Cross-polytope	$\{\pm e_k\}_{k=1}^d$; all nonantipodal pairs	$1/2$	$d-1$	1	d	$d-1$	$d(d-1)/2$
Hypercube	$\{\pm 1/\sqrt{d}\}^d$; cube edges	$(d-1)/2$	$d(d-1)/2$	$2/d$	2	$\binom{d}{2}$	$d-1$

For each family, (5.3) holds with $c_* = d(d-1)/r_{\max}$. Consequently,

$$LS_2 = -d(2-\ell_*)S_2,$$

where ℓ_* is the common active loss. These examples establish sharpness in every ambient dimension; no extrapolation from fixed-dimensional numerics is used.

8.2 Platonic examples in $d = 3$

With uniform masses, natural shortest-edge graphs, and their exact equality rates, the tetrahedron, octahedron, cube, icosahedron, and dodecahedron all satisfy

$$\mathfrak{D}_2 r_{\max} = 6, \qquad E_{\text{sample}} = \{0\}.$$

The tetrahedral, octahedral, and cubical coefficient spaces have alias dimensions 2, 3, and 2, respectively. Thus nonzero trace-free quadratic forms may vanish on every node even at frontier equality. The icosahedral and dodecahedral quadratic sampling maps are injective.

8.3 What the examples do and do not show

The examples prove attainment and demonstrate that sampling aliases are compatible with equality. They do not classify all extremizers. Reversible blowups, covers, long-chord symmetric shells, and higher-dimensional equality families prevent such a classification without further hypotheses. Separately, nonregular weighted tangent tight frames show that even local equality does not force a regular vertex figure.

9. Rejected claims and deterministic counterexamples

The following statements are false or unsupported and are excluded from the paper.

Rejected claim	Obstruction retained in the proof package	Correct replacement
The Frobenius norm on $\mathrm{Sym}_0(d)$ can replace the sampled quotient norm.	Nontrivial K_X in the simplex, cross-polytope, cube, and Platonic examples.	Work on $Q_X = V/K_X$ with metric (2.2).
Frontier equality reproduces H_2 exactly.	At equality $R_2 = c_* S_2$ with $c_* > 0$, while $E_{\text{sample}} = \{0\}$.	Equality minimizes the sampled residual under the positivity–rate constraint.
Trace saturation alone implies full frontier equality.	Unequal-rate and repeated-node exact fixtures separate the trace step from the rate and variance steps.	Require all three conditions in (5.1).
Local positive row stencils automatically assemble into a reversible generator.	Cycle incompatibility and detailed-balance failures.	Impose the global Markov, Gram, and Kolmogorov conditions of Theorem 5.3.
Every connected equality generator is Platonic.	Exact blowups, covers, long-chord icosahedral shells, and all-dimensional families.	Use Corollary 5.5 only under its explicit support and convexity hypotheses.
Small weighted defect implies uniform vertexwise closeness.	An order-one defect can be hidden at a vertex of arbitrarily small stationary mass.	Retain (6.9) or assume $w_{\min} > 0$.
Small global shell variance controls every active edge.	An edge of arbitrarily small directed conductance probability may carry order-one loss defect.	Assume $p_{ij} \geq \kappa$, retain the vertex-mass dependence, and use $w_{\min} > 0$ for a uniform all-edge statement.
Local edge control propagates globally without a graph parameter.	Long paths, cycles, and small-gap graphs accumulate variation.	Retain the $\kappa, w_{\min}$ margins and the applicable path, resistance, gap, or congestion dependence.
Small covariance defect alone yields stable eigenvectors or frame repair.	Isotropic targets, nearly singular tangent frames, antipodal collapse, and zero-loss bridges.	Add spectral-gap, tangent-frame, shell, overlap, and surjectivity margins as appropriate.

Rejected claim	Obstruction retained in the proof package	Correct replacement
A fitted $O(h^2)$ slope proves the matching upper theorem.	Any finite prefix is compatible with different asymptotics and does not certify exactness or positivity at all levels.	Use the all-orders symbolic, Cauchy, recurrence, and positivity proof of Theorem 7.2.
Positivity universally forbids higher-order graph Laplacians.	Signed high-order graph formulas use cancellation outside the present class.	State only the scoped product theorem (4.3).

10. Consequences deliberately outside the flagship theorem

No generator-lift, semigroup comparison, transport-cost improvement, or physical-model performance theorem is claimed here. A signed construction may illustrate that leaving the positive cone changes the obstruction, but it does not strengthen (4.3) unless its normalization and comparison class are made identical. Such material should appear only in a later paper or a clearly separated discussion after independent proof and model audits.

Appendix A. Proof dependency graph

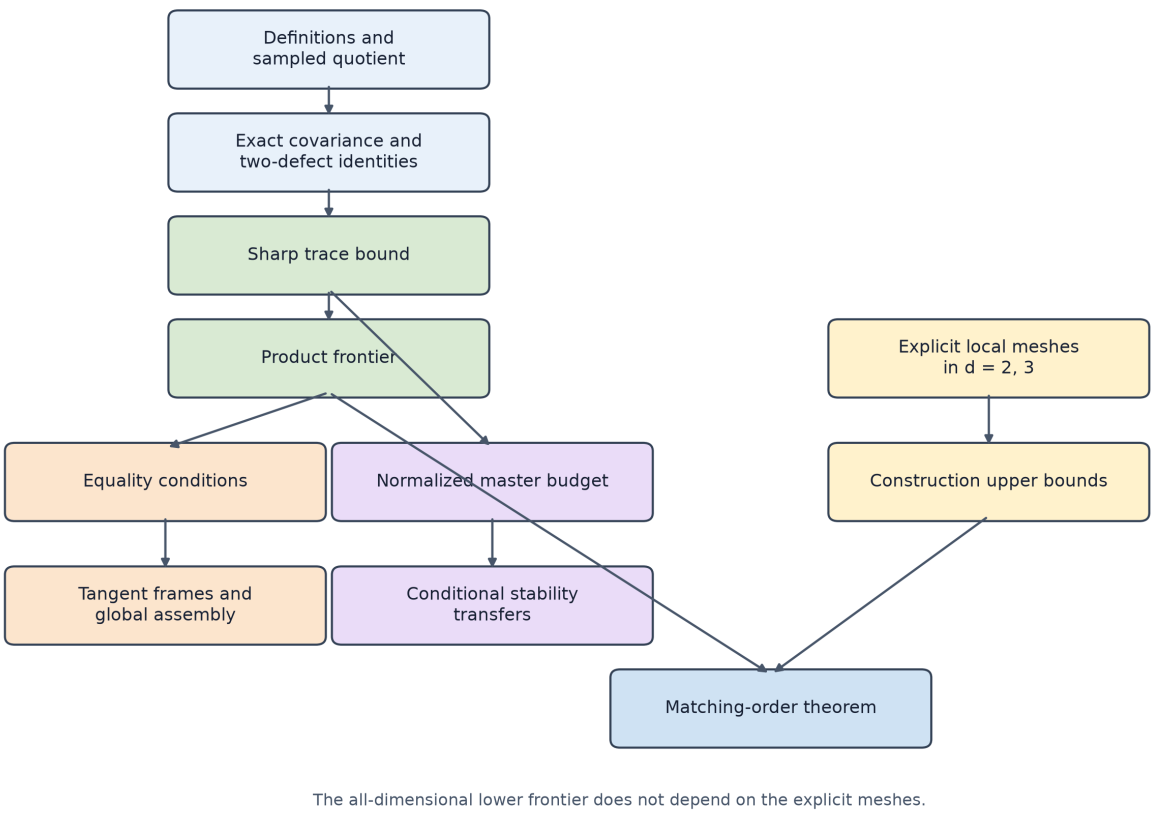

Figure 1: Proof dependency graph. The lower frontier is independent of the explicit meshes; the matching result combines the frontier with the construction upper bounds.

The all-dimensional lower frontier does not depend on the constructions. The matching corollaries use it only after the independent `d=2` or `d=3` upper family has verified positivity, exact coordinates, and the applicable rate cap. No construction is asserted for `d>3`.

Appendix B. Theorem-to-source-to-test-to-registry map

The proof source is controlling. A test source supplies falsification and regression evidence but is not a substitute for proof.

Manuscript result	Proof source	Test or audit source	Claim-registry entry	Status in this draft
Proposition 2.1, sampled quotient, Gram pencil, sampled exact-space dimension	afp_barrier_gate1/pure_math/covariance/P1A_QUADRATIC_FIDELITY_FOUNDATION.md, Sections 4–6	afp_barrier_gate1/pure_math/covariance/p1a_quadratic_fidelity_audit.py	P1A-QUOT; P1A-GRAM	ordinary proof with Lean-checked finite identities
Theorem 3.1, exact covariance and two-defect identities	same P1A source, Sections 2–3	same P1A audit	P1A-2DEF	ordinary proof with Lean-checked finite identities
Proposition 3.2, weighted adjoints and traces	same P1A source, Sections 4–6	same P1A audit	P1A-GRAM	ordinary proof with Lean-checked finite identities
Theorem 4.1, trace inequality and sharp product	afp_barrier_gate1/pure_math/covariance/P1B_SHARP_QUADRATIC_DEFECT_BOUND.md, Sections 3–13	afp_barrier_gate1/pure_math/covariance/p1b_sharp_quadratic_defect_audit.py	P1B-TRACE; P1B-SHARP	ordinary proof with Lean-checked lower-bound algebra
Theorem 5.1, equality characterization	same P1B source, Section 11; P1C source for geometric equivalences	P1B and P1C audits	P1B-EQUALITY; P1C-GLOBAL	ordinary proof; selected local algebra Lean-checked
Theorem 5.2, tangent block and tight-frame equivalence	afp_barrier_gate1/pure_math/covariance/P1C_EQUALITY_GEOMETRY.md	afp_barrier_gate1/pure_math/covariance/p1c_equality_geometry_audit.py	P1C-LOCAL	ordinary proof; selected equivalences Lean-checked
Theorem 5.3, global reversible assembly	same P1C source	same P1C audit	P1C-GLOBAL	ordinary mathematics; no end-to-end Lean theorem
Proposition 5.4 and Corollaries 5.5–5.6, antipodal, scoped classification, and global-moment consequences	same P1C source	same P1C audit	P1C-GLOBAL; P1C-CLASS; P1C-PLATONIC	ordinary mathematics; Corollary 5.5 uses the cited classical classification
Theorem 6.1, normalized stability budget	afp_barrier_gate1/pure_math/covariance/P1D_QUANTITATIVE_STABILITY.md, Sections 1–3	afp_barrier_gate1/pure_math/covariance/p1d_quantitative_stability_audit.py	P1D-MASTER	ordinary proof with the printed finite consequences Lean-checked
Corollary 6.2, exceptional mass and pointwise transfer	same P1D source, Sections 4–5	same P1D audit	P1D-VERTEX	ordinary proof with matching Lean inequalities

Manuscript result	Proof source	Test or audit source	Claim-registry entry	Status in this draft
Conditional transfers discussion after Corollary 6.2	same P1D source, Sections 6–10	same P1D audit	P1D-EDGE; P1D-GRAPH; P1D-COV; P1D-QUOTIENT; P1D-FRAME	not a numbered theorem; no global embedding-stability claim
Section 8, all-dimensional and Platonic exact families and aliases	P1C source	P1C audit	P1C-FAMILIES; P1C-PLATONIC; P1C-ALIAS	ordinary exact mathematics
Section 9, unrestricted-classification rejection	P1C and P1D counterexample sections	P1C and P1D audits	P1C-UNRESTRICTED and caveats attached to P1D entries	rejected claims retained
Theorem 7.1, regular-polygon matching family	Section 7.1, direct Fourier proof	p1e_asymptotic_family_audit.py; QuadraticFidelityConstruction.lean	P1E-POLYGON	ordinary proof for `d=2`; finite recurrences Lean-checked
Theorem 7.2, adaptive-ring mesh, positivity and exact H_0,H_1	docs/publication_program/P1E_SHORT_GAP_S2_CONSTRUCTION.md, Sections 1–8	exact-rational certificate and standalone verifier; derivation audits listed below	P1E-RING	computer-assisted exact-rational proof for `d=3`
Theorem 7.2, sampled quotient multiplier and matching constants	P1E proof source, Section 9	certificate verifier; proof audit; hostile Cauchy audit	P1E-ROW; F-CONSTRUCT	computer-assisted exact-rational proof for `d=3`
Open problem 7.3, support-preserving perturbations	P1E source, Section 10, and retained diagnostic calculations	p1e_no_guard_ring_independent_audit.py	P1E-ROBUST	open; no perturbation radius or robustness constants are claimed

The claim registry is THEOREM_REGISTRY.md. Any change in theorem strength must first be reconciled with the proof source and then recorded in that registry; a passing test alone cannot promote a claim.

Appendix C. Falsification protocol

The computational appendices serve four limited purposes:

1. verify finite tensor identities and sign conventions on deterministic exact examples;
2. expose sampling aliases, leakage, and weighted-adjoint mistakes;
3. retain deterministic failure cases for omitted hypotheses; and
4. check explicit constants and extreme-scale conditioning with exact, algebraic, or outward-rounded interval arithmetic where appropriate.

The required audit programs are:

- p1a_quadratic_fidelity_audit.py for quotient, covariance, weighted Gram, and alias identities;
- p1b_sharp_quadratic_defect_audit.py for sharpness, degeneracy, sign, and normalization fixtures;

- p1c_equality_geometry_audit.py for tangent blocks, all-dimensional extremizers, Platonic exactness, aliases, antipodal branches, blowups, long-chord shells, weighted frames, and cycle compatibility; and
- p1d_quantitative_stability_audit.py for small masses, small active edge probabilities, long paths, small gaps, singular frames, sampling gaps, aliases, and certified extreme-scale inequalities;
- p1e_short_gap_symbolic_matrix_audit.py, p1e_short_gap_cauchy_guard_audit.py, and p1e_short_gap_cauchy_hostile_audit.py for the literal transition matrix and its uniform all-orders guard;
- p1e_short_gap_polar_guard_audit.py, p1e_short_gap_proof_audit.py, and p1e_no_guard_ring_independent_audit.py for the first row, exact constants, recurrence, and reachable-domain margins; its perturbation arithmetic is a retained candidate diagnostic, not theorem evidence; and
- p1e_short_gap_family_audit.py and p1e_short_gap_referee_audit.py for finite generator reconstruction and deterministic failed mutations. The six alternative-route audits are retained separately and cannot be read as a proof by exhaustion.

No numerical rank threshold is used in a theorem. Exact symbolic minors or algebraic identities certify the ranks quoted in Section 8. Theorem 7.2 is all-orders; its finite generator output and convergence measurements are regressions only.

Appendix D. Formalization

The finite algebraic core is formalized in the following Lean sources:

Formal source	Mathematical scope
AFPBarrier/QuadraticFidelityFoundation.lean	weighted finite identities, sampling quotient foundations, covariance decomposition
AFPBarrier/QuadraticFidelityLowerBound.lean	trace comparison and sharp lower-bound algebra
AFPBarrier/QuadraticEqualityGeometry.lean	division-free tangent blocks and guarded normalized-frame identities
AFPBarrier/QuadraticFidelityStability.lean	normalized master-budget algebra and finite consequences
AFPBarrier/QuadraticFidelityConstruction.lean	shared-stress force and detailed-balance identities, moment normalization, four-point connectors, rate conversion, and regular-polygon residuals

The formalization is a second proof surface for the stated finite identities. It does not by itself formalize every geometric classification argument, external-priority statement, adaptive-ring schedule, or analytic Cauchy guard. The manuscript must not label an unformalized conclusion as machine-checked merely because a nearby algebraic lemma is formalized.

Appendix E. Reproducibility record

From the repository root, the deterministic theorem audits may be run with:

```
python afp_barrier_gate1/pure_math/covariance/p1a_quadratic_fidelity_audit.py
python afp_barrier_gate1/pure_math/covariance/p1b_sharp_quadratic_defect_audit.py
python afp_barrier_gate1/pure_math/covariance/p1c_equality_geometry_audit.py
python afp_barrier_gate1/pure_math/covariance/p1d_quantitative_stability_audit.py
python afp_barrier_gate1/pure_math/covariance/p1e_short_gap_symbolic_matrix_audit.py
python afp_barrier_gate1/pure_math/covariance/p1e_short_gap_cauchy_guard_audit.py
python afp_barrier_gate1/pure_math/covariance/p1e_short_gap_cauchy_hostile_audit.py
python afp_barrier_gate1/pure_math/covariance/p1e_short_gap_polar_guard_audit.py
python afp_barrier_gate1/pure_math/covariance/p1e_short_gap_proof_audit.py
python afp_barrier_gate1/pure_math/covariance/p1e_no_guard_ring_independent_audit.py
python afp_barrier_gate1/pure_math/covariance/p1e_short_gap_family_audit.py
python afp_barrier_gate1/pure_math/covariance/p1e_short_gap_referee_audit.py
```

The Lean surface may be checked from afp_barrier_gate1 using the repository's pinned toolchain and build command.

Release-specific workflow identifiers, continuous-integration run identifiers, commit hashes, environment captures, and artifact checksums belong only in a separate reproducibility manifest accompanying a tagged release. They are not part of any hypothesis or mathematical argument and must not appear in Sections 1–10.

The external-source bibliography for the priority audit is docs/publication_program/p1f_manuscript/priority_sources.bib. Every imported theorem must be accompanied by the source-specific hypothesis-transfer note in PRIORITY_AND_HOSTILE_REFEREE_AUDIT.md; the present P1A–P1E proofs are self-contained and use those sources for context rather than as hidden proof lemmas.

Appendix F. Limitations and future work

1. The matching construction is proved only in ambient dimensions two and three. The sharp frontier, equality theory, and quantitative stability are all-dimensional, but no matching local family is claimed for `d>3`.
2. The stability theorem is fundamentally weighted. Uniform pointwise rigidity requires a lower stationary mass; uniform edgewise rigidity requires a lower active conductance probability.
3. Graphwide rigidity depends on a quantitative connectivity parameter. Sampling-quotient scalarity depends on a lower sampling-frame bound, and tangent-frame repair depends on tangent conditioning and nondegeneracy margins.
4. Equality is not a classification by a short list of symmetric polytopes. The scoped three-dimensional convex classification has explicit hypotheses, while blowups, covers, long-chord shells, and higher-dimensional equality families remain; nonregular local frames separately obstruct local regularity.
5. Sampling aliases can occur for small or symmetric node sets. The theorem controls the genuinely sampled quotient and does not reconstruct coefficients in K_X.
6. No transport, physical-model, or application-level performance consequence follows from the harmonic operator norm alone. Such claims require separate models, normalizations, and audits.
7. Signed higher-accuracy graph operators are outside the positive reversible class. A future comparison should specify a common rate or cost constraint before drawing conclusions about the benefit of leaving the positive cone.
8. The candidate `d=3` support-preserving perturbation route fixes ring counts, phases, masks, horizontal jumps, pole/equator data, and reflection. It remains an open certification problem, supplies no theorem, perturbation radius, or constants, and does not cover arbitrary node motion or independent longitude perturbations.

The immediate construction question is now higher-dimensional: whether a different local compiler can produce the matching order for `d>3` with shared positive conductances and a quantitative global feasibility margin. A broader robustness result in `d=3` would likewise require a uniform right inverse for the full shared six-moment edge system.

Appendix G. Related-theorem and hypothesis-transfer matrix

The table distinguishes external results from the theorems proved here. The long-form line-by-line convention audit, including primary-source theorem locations, is PRIORITY_AND_HOSTILE_REFEREE_AUDIT.md.

Source and variables	External hypotheses	External conclusion used	Overlap	New distinction and transfer guard
Babecki–Thomas: graph eigenspaces, node subsets, quadrature weights	connected regular unweighted graph; normalized-adjacency ordering	graphical designs correspond through Gale duality to eigenpolytope faces, with support consequences	finite eigenspaces and positive weights	their weights average a fixed graph; w_i and γ_{ij} here define the operator. No spherical eigenmap, rate cap, or quadratic frontier transfers

Source and variables	External hypotheses	External conclusion used	Overlap	New distinction and transfer guard
Babecki–Shiroma: positive edge-weighted graphs and eigenpolytopes	connected graph; symmetric combinatorial Laplacian in uniform Euclidean metric	broad eigenspace and eigenpolytope universality for positive weighted graphs	prescribed eigenspaces with positive edges	universality need not preserve unit spherical geometry, locality, degree, nonuniform mass, target eigenvalue, or H_2 residual; flip the positive-semidefinite sign and retain the mass matrix
Steinerberger (Steinerberger 2021): manifold eigenfunctions, n quadrature nodes	compact boundaryless manifold; nonnegative quadrature exact on an initial spectral segment	asymptotic cardinality limitation for positive quadrature	positivity and continuous harmonic modes	integration is not a graph-operator identity. Keep manifold dimension $d-1$, multiplicity, volume measure, and the asymptotic $o(n)$ term; no finite frontier follows
Izmestiev–Lam: geodesic spherical triangulation, c_{ij}, d_i	closed triangulation; nonnegative Delaunay edge coefficients	their normalized spherical Laplacian is reversible and satisfies $\Delta_s p = -2p$	local positive reversible coordinate-exact operators on $\mathbb{S}^2$	this paper does not claim to introduce that object. Their theorem supplies neither the sampled quotient/frontier nor the adaptive-ring H_2 and mesh constants; conversion requires $w_i = d_i / \sum d$, $\gamma_{ij} = c_{ij} / \sum d$
Seibold: Euclidean point cloud and row stencil coefficients	polynomial moment equations; local geometric cone conditions	sparse positive Poisson stencils and feasibility criteria	positive local moment cones	a rowwise stencil is generally nonsymmetric and gives no shared γ_{ij} or stationary masses. Curvature, global reconciliation, and a strict margin must be proved separately
Babecki–Steinerberger–Thomas: base graph and first k eigenpairs	connected positive weighted graph; candidate is a reweighted spanning subgraph	eigenpair-preserving Laplacians form a polyhedral/spectrahedral slice	positive eigenpair preservation	here the geometric coordinate module is prescribed, aliases are quotiented, and the next continuous shell is measured under $r_{\max}$. With nonuniform w, the comparison is a generalized eigenproblem
García Trillos–Gerlach–Hein–Slepčev: random manifold sample and kernel bandwidth	i.i.d. density bounds, compact manifold geometry, kernel and bandwidth conditions	high-probability eigenvalue/eigenvector convergence to a weighted Laplace–Beltrami operator	local positive graph Laplacians approaching a continuum operator	probabilistic spectral convergence does not give finite exact H_1, bounded degree, shared correction, or the deterministic $O(h^2)$ quotient bound. Their bandwidth is not silently identified with this fill parameter

Source and variables	External hypotheses	External conclusion used	Overlap	New distinction and transfer guard
Martin–Tanaka: association scheme adjacency algebra	finite commutative scheme; uniform counting metric	primitive idempotents and nonnegative Krein parameters organize products and embeddings	symmetric examples and harmonic-module products	one must first prove that L lies in the Bose–Mesner algebra and identify its idempotents with sampled spherical harmonics. Unequal masses and aliases are not automatic
Bannai–Bannai: finite spherical sets and polynomial moments	usually uniform spherical t-design measure	harmonic-moment characterization and design bounds	common symmetric node sets and frame moments	design exactness is a measure identity, not a generator. It implies neither reversible local edges, sampled injectivity, nor L-invariance; frontier equality has positive residual, not exact H_2
Ahrens–Beylkin (Ahrens and Beylkin 2009): icosahedral orbits and quadrature weights	rotational invariance; moment equations for targeted polynomial spaces	rotationally invariant spherical quadratures	symmetry-orbit reductions	quadrature weights are not conductances or stationary masses. Symmetry preserves a generator constraint only after support and orientation invariance are proved
Yoon: graph powers/walks and m-Laplacian coefficients	simple graph; signed higher-order finite-difference-inspired operator	high formal order on cycles using alternating coefficients for $m \geq 2$	graph-Laplacian accuracy	the signed class lies outside $\gamma \geq 0$. The present theorem is only the sampled degree-two residual/rate frontier for positive reversible coordinate-exact generators
Benedetto–Fickus (Benedetto and Fickus 2003): unit vectors and frame operator	finite unit-norm frames in a Hilbert space	unit-norm tight frames minimize the frame potential	tangent tight-frame equality language	tight frames themselves are established theory. The new statement is their forced appearance together with common loss/rate and global detailed-balance compatibility in equality for this frontier

No row is imported as a proof of Theorems 3.1, 4.1, 5.1–5.3, 6.1, or 7.1–7.2. External results supply context and established terminology; every normalization used in the new theorem hierarchy is derived internally.

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Bibliographic data for the sources cited in the priority discussion are in priority_sources.bib. Journal versions are used when available, and the related-work discussion retains the convention-transfer notes needed to keep the comparison classes distinct.

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